ANOVA Methods: statistics, psychometrics, and APA reporting for psychology in Python

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Summary

ANOVA Methods is a family of three Python packages covering the analyses that appear in most quantitative psychology reports. `psystats` provides descriptive statistics, group comparison tables, inferential tests with effect sizes, regression, and epidemiological risk measures. `psymetrics` provides reliability analysis, factorability checks, exploratory factor analysis, and confirmatory factor analysis with structural equation models. `psyreport` renders the results of the other two as APA 7 text, LaTeX tables, and Word documents.

The three packages share one convention. Every analysis returns a lightweight result object carrying a `kind` tag, a dictionary of raw estimates, and an optional table. Printing the object gives an R style console summary, and passing it to `psyreport.report()` gives APA 7 prose. Dispatch happens on the `kind` tag, so `psyreport` renders a result without importing the package that produced it. One computation therefore feeds a console summary, a LaTeX table, and a Word document without being recomputed.

Statement of need

Psychologists working in R use a small set of packages that pair a concise interface with output shaped for reporting. These include `arsenal` and `tableone` for group comparison tables (Heinzen et al., 2021; Yoshida & Bartel, 2022), `epitools` for risk measures (Aragon, 2020), `psych` for reliability and exploratory factor analysis (Revelle, 2024), `lavaan` for structural equation models (Rosseel, 2012), and `apaTables` and `report` for formatted output (Makowski et al., 2023; Stanley, 2021).

Python provides the estimation machinery for these analyses through `numpy` (Harris et al., 2020), `pandas` (McKinney, 2010), `scipy` (Virtanen et al., 2020), `statsmodels` (Seabold & Perktold, 2010), `factor_analyzer` (Biggs et al., 2021), and `semopy` (Meshcheryakov & Igolkina, 2020). Those libraries expose general purpose interfaces. Moving from them to a reportable table takes considerable boilerplate, because the analyst selects a test appropriate to each variable, computes the matching effect size and confidence interval, and formats the result by hand. The workflow that makes the R packages attractive, in which a data frame becomes an APA table in one call, has no direct Python equivalent.

ANOVA Methods supplies that workflow. It wraps established estimators behind verbs that mirror their R analogues, so a researcher who already knows the R tools incurs little translation cost. `table1(df, group="country")` builds a group comparison table and selects a test per variable, `riskratio(df, exposure=..., outcome=...)` returns a risk ratio with its confidence interval, and `alpha(items)` returns coefficient alpha with item level statistics. A user coming from Python obtains effect sizes, confidence intervals, and automatic test selection without assembling them by hand.

The intended users are researchers, students, and instructors in psychology and allied fields who work in Python and need reportable output for routine analyses. A documented teaching dataset ships inside `psystats`, so the examples in the documentation run without any download. It contains responses from 1,957 undergraduate students in Spain, Portugal, and Brazil on the Beck Depression Inventory and the Beck Anxiety

Inventory, together with cyber victimization, cyber aggression, and emotion regulation scales (Afonso Junior et al., 2020).

Software description

`psystats` covers four areas. Descriptives and tables are handled by `describe`, `freq`, `corr_matrix`, which reports a correlation matrix with significance stars, and `table1`. For each variable `table1` selects a Welch t test, a one way analysis of variance, a Mann-Whitney test, a Kruskal-Wallis test, a χ^2 test, or Fisher's exact test according to the variable type, the number of groups, and a Shapiro-Wilk screen for normality. Inferential tests with effect sizes are handled by `ttest`, which reports Cohen's d, `anova`, which reports η^2 and ω^2, and `chisq`, which reports Cramér's V. Regression is handled by `linreg`, which reports standardized coefficients and model fit, and `logreg`, which reports odds ratios with confidence intervals. Risk factors are handled by `riskratio`, `oddsratio`, and `attributable_risk`, the last returning the risk difference with its confidence interval, the attributable risk percent, the population attributable risk and its percent, and the number needed to treat.

`psymetrics` covers reliability and latent variable models. `alpha` returns coefficient alpha with the Feldt confidence interval, the mean inter item correlation, and a per item table of corrected item total correlations and alpha if dropped. `kmo` and `bartlett` provide the two standard factorability checks. `efa` runs exploratory factor analysis with selectable estimators and rotations. `cfa` fits confirmatory factor models and structural equation models from lavaan style syntax, so a model specification transfers from R unchanged.

`psyreport` renders results as APA 7 text through `report`, and exports tables through `to_latex` and `to_docx`. It currently renders 15 result types, which covers everything `psystats` emits and all of `psymetrics` except the two factorability checks.

Example

```
from psystats import load_mapfre, logreg, anova
from psymetrics import alpha
from psyreport import report

df = load_mapfre()
bdi = df[[f"bdi_{i}" for i in range(1, 22)]]
print(report(alpha(bdi)))

df["depressed"] = (df["bdi_sum"] >= 14).astype(int)
print(report(logreg(df, outcome="depressed", predictors=["bai_sum"])))
print(report(anova(df, dv="bdi_sum", group="country")))
```

```
Cronbach's α = .89, 95% CI [.89, .90], based on 21 items (N = 1905).

Logistic regression predicting depressed.
McFadden pseudo-R² = .19, N = 1952.

bai_sum: OR = 1.15, 95% CI [1.13, 1.16], p < .001.

F(2, 1946) = 8.85, p < .001, η² = .01.
```

Quality control

An automated test suite of 50 tests runs on every change. Estimates are checked against independent references rather than against the packages' own internals. Test statistics, confidence intervals, and regression coefficients are compared with `scipy` and `statsmodels`, effect sizes are compared with closed form identities, and the fit indices returned by `cfa` are compared with `semopy` computed directly on the Holzinger and Swineford data. The documentation for `psystats` is generated by executing every example against the bundled dataset, so the output shown in it is produced by the code rather than transcribed.

Availability

The three packages are published on PyPI and installable with `pip install psystats psymetrics psyreport`. Confirmatory factor analysis requires `semopy`, which is kept as an optional extra so that the core installs without a compiler toolchain. The source is developed in one repository at https://github.com/anovabr/anova-methods under the MIT license.

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Software repository: https://github.com/anovabr/anova-methods · License: MIT · Archived releases and the citable DOI are listed in `CITATION.cff`.